Unit 1

1. Introduction to AI

Module 1.6

1.6 Hardware for AI-Based Systems

Tutorial 1.6

1.6 Hardware for AI-Based Systems

AI-1.6.1 K2 Compare the choices available for hardware to implement AI-based systems.

Hardware for AI-Based Systems

- A variety of hardware is used for ML model training and model implementation.
- A model that performs speech recognition may run on a low-end smartphone, although access to the power of cloud computing may be needed to train it.
- A common approach used when the host device is not connected to the internet is to train the model in the cloud and then deploy it to the host device.
- ML typically benefits from hardware that supports the following attributes:
  - Low-precision arithmetic: This uses fewer bits for computation (e.g., 8 instead of 32 bits, which is usually all that is needed for ML).
  - The ability to work with large data structures (e.g., to support matrix multiplication).
  - Massively parallel (concurrent) processing
- General-purpose CPUs provide support for complex operations that are not typically required for ML applications and only provide a few cores.
- As a result, their architecture is less efficient for training and running ML models when compared to GPUs, which have thousands of cores and which are designed to perform the massively parallel but relatively simple processing of images.
- As a consequence, GPUs typically outperform CPUs for ML applications, even though CPUs typically have faster clock speeds.
- For small-scale ML work, GPUs generally offer the best option.

Hardware for AI-Based Systems Cont.

- Some hardware is specially intended for AI, such as purpose-built Application-Specific Integrated Circuits (ASICs) and System on a Chip (SoC).
- These AI-specific solutions have features such as multiple cores, special data management and the ability to perform in-memory processing.
- They are most suitable for edge computing, while the training of the ML model is done in the cloud.
- Hardware with specific AI architectures like:
  - Neuromorphic processors, which do not use the traditional von Neumann architecture, but rather an architecture that loosely mimics brain neurons.

Examples of AI hardware providers

NVIDIA

- NVIDIA provides a range of GPUs and AI-specific processors

Product	Description	Use Case Highlights
NVIDIA H100 / H200 GPUs	Flagship data center GPUs with Transformer Engine for LLMs	Training massive generative AI models
NVIDIA Grace Blackwell (GB200)	AI superchip combining CPU & GPU for extreme performance	Scientific computing, enterprise AI
NVIDIA DGX Systems	Turnkey AI supercomputers with full-stack integration	Research labs, enterprise AI infrastructure
NVIDIA RTX AI PCs	Consumer-grade GPUs with AI acceleration (e.g., RTX 4060, 4080)	Generative AI, gaming, content creation
Jetson Orin Nano Super Kit	Affordable edge AI platform with up to 67 TOPS performance	Robotics, IoT, vision transformers

Examples of AI hardware providers

Google

- Google has developed application-specific integrated circuits for both training and inferencing.
- Google TPUs (Cloud Tensor Processing Units) can be accessed by users on the Google Cloud, whereas the Edge TPU is a purpose-built ASIC designed to run AI on individual devices.

TPU Generation	Key Features & Innovations	Use Case Focus
TPU v1–v4	Designed for training and inference of deep learning models; used in Google Cloud	General-purpose AI workloads
Trillium (TPU v6)	Delivered **4x speed** over previous gen, optimized for **Gemini 2.0** models	Large-scale training & inference
Ironwood (TPU v7)	Purpose-built for the “age of inference”; scales to **9,216 chips** with **42.5 Exaflops**	High-performance inferential AI

Examples of AI hardware providers

Intel

- Intel provides Nervana neural network processors (newer platforms like Habana Gaudi and Core Ultra NPUs) for deep learning (both training and inferencing) and Movidius Myriad vision processing units for inferencing in computer vision and neural network applications.
- Neuromorphic Computing (Loihi Series)
  - Intel’s Loihi 2 is a cutting-edge neuromorphic processor that mimics the brain’s spiking neural networks
    - Supports up to 1 million neurons and 120 million synapses
    - Enables event-driven computation for ultra-efficient edge AI
    - Powers systems like Hala Point, the world’s largest neuromorphic system with 1.15 billion neurons
    - Comes with Lava, an open-source framework for neuro-inspired applications
- AI Accelerators & Data Center Chips
  - Intel offers several high-performance solutions for training and inference:
    - Habana Gaudi2: Deep learning accelerator optimized for generative AI and LLMs
    - Intel Xeon Scalable Processors: With built-in AI acceleration for data prep and model deployment
    - Intel Xeon D & W: Tailored for edge and workstation AI workloads

Examples of AI hardware providers

Mobileye

- Mobileye produce the EyeQ family of SoC devices to support complex and computationally intense vision processing. These have low power consumption for use in vehicles.

Hardware Component	Description & Capabilities
EyeQ™ SoC Family	Custom system-on-chips designed for ADAS and autonomous driving. Latest version: **EyeQ6H**
Imaging Radar	High-resolution radar tech that complements cameras, addressing their blind spots
Mobileye Drive™ ECU	Modular compute unit for full autonomy which is built on EyeQ6H

Examples of AI hardware providers

Apple

- Apple produce the Bionic chip for on-device AI in iPhones

Chip / Platform	Description & Capabilities
M3 Ultra	Apple’s most powerful chip yet, with a 32-core CPU, 80-core GPU, and 32-core Neural Engine
Baltra (AI Server Chip)	In development with Broadcom, designed for data centers and Apple Intelligence workloads
Apple Neural Engine (ANE)	Integrated into A-series and M-series chips and optimized for on-device ML tasks like image recognition
Sotra, Komodo, Borneo	Upcoming SoCs for future iPads, Macs, and wearables. It is expected to enhance AI capabilities

Examples of AI hardware providers

Huawei

- Huawei’s Kirin 970 chip (Legacy) for smartphones has built-in neural network processing for AI

Product	Description & Capabilities
Ascend 910C	Huawei’s top-tier AI processor, used in large-scale training systems
CloudMatrix 384	AI supercomputing system with 384 Ascend 910C chips; rivals NVIDIA’s GB200 NVL72
CANN Toolkit	Compute Architecture for Neural Networks—Huawei’s open-source alternative to NVIDIA CUDA

THANK YOU!!!